

Abbreviations
$\sim X_1 = ((\text{TLS12}, \sim M_6, \text{RSA_cipher}, \text{extensions}), \sim M_9, \text{ServerHelloDone}) = ((\text{TLS12}, \text{rs_4}, \text{RSA_cipher}, \text{extensions}), \text{sign}(\text{pk}(\text{skS_2}), \text{caSKey_1}), \text{ServerHelloDone})$
$\sim M_11 = \text{RSAenc}(\text{PMS_4}, \text{pk}(\text{skS_2}))$
$\sim M_12 = \text{CCS}$
$\sim M_13 = \text{PRF}(\text{PRF}(\text{PMS_4}, \text{master_secret}, (\text{rc_4}, \text{rs_4})), \text{client_finished}, \text{Hash}(((\text{TLS12}, \text{rc_4}, \text{RSA_cipher}, \text{extensions}), \text{sign}(\text{pk}(\text{skS_2}), \text{caSKey_1}), \text{RSAenc}(\text{PMS_4}, \text{pk}(\text{skS_2}))))))$
$\sim X_2 = (\sim M_11, \text{CCS}, \sim M_13) = (\text{RSAenc}(\text{PMS_4}, \text{pk}(\text{skS_2})), \text{CCS}, \text{PRF}(\text{PRF}(\text{PMS_4}, \text{master_secret}, (\text{rc_4}, \text{rs_4})), \text{client_finished}, \text{Hash}(((\text{TLS12}, \text{rc_4}, \text{RSA_cipher}, \text{extensions}), \text{sign}(\text{pk}(\text{skS_2}), \text{caSKey_1}), \text{RSAenc}(\text{PMS_4}, \text{pk}(\text{skS_2}))))))$
$\sim M_14 = \text{CCS}$
$\sim M_15 = \text{PRF}(\text{PRF}(\text{PMS_4}, \text{master_secret}, (\text{rc_4}, \text{rs_4})), \text{server_finished}, \text{Hash}(((\text{TLS12}, \text{rc_4}, \text{RSA_cipher}, \text{extensions}), \text{sign}(\text{pk}(\text{skS_2}), \text{caSKey_1}), \text{RSAenc}(\text{PMS_4}, \text{pk}(\text{skS_2}))))))$
$\sim X_3 = (\text{CCS}, \sim M_15) = (\text{CCS}, \text{PRF}(\text{PRF}(\text{PMS_4}, \text{master_secret}, (\text{rc_4}, \text{rs_4})), \text{server_finished}, \text{Hash}(((\text{TLS12}, \text{rc_4}, \text{RSA_cipher}, \text{extensions}), \text{sign}(\text{pk}(\text{skS_2}), \text{caSKey_1}), \text{RSAenc}(\text{PMS_4}, \text{pk}(\text{skS_2}))))))$

A trace has been found.

